

Chapter 2.

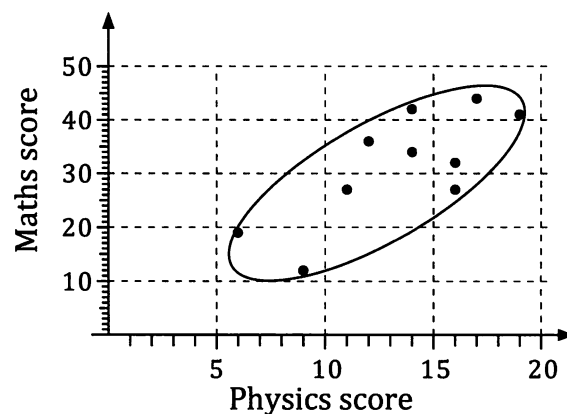
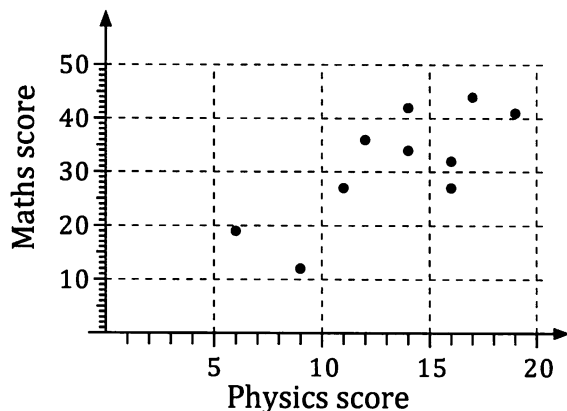
Bivariate data – further analysis.

In the previous chapter, after the work involving *categorical data*, we investigated whether *relationships*, *associations* or *correlation* existed between two *numerical* variables. The method was to plot a scattergraph of paired values to see if the scatter of points suggested some linear, or other, model could satisfactorily summarise what was going on. I.e we considered what **form** (linear/non linear) any association might take. Let us now try to be more *descriptive* and *quantitative* about situations for which a linear model seems appropriate.

Suppose that ten students took both a physics test and a mathematics test and the scores they obtained were as shown in the following table:

Physics Score	6	9	11	12	14	14	16	16	17	19
Maths Score	19	12	27	36	42	34	32	27	44	41

The scattergraph for these scores is shown below left. The fact that the points tend to form an oval shape from lower left to upper right, as shown below right, indicates that for these students, those who did well in the physics test tended to do well in the mathematics test, and those who scored the lower marks in the physics test tended to score the lower marks in the mathematics test



We say that the scattergraph suggests that a **positive correlation** exists between the physics scores and the maths scores.

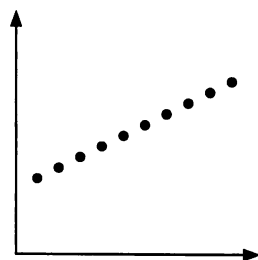
If a positive correlation exists between two variables then as one variable increases the other variable increases. Our "line of best fit" will have a positive gradient.

If instead the points had formed an oval shape from upper left to lower right that would have suggested a **negative correlation** existed between the variables.

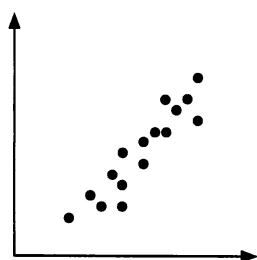
If a negative correlation exists between two variables then as one variable increases the other variable decreases. Our "line of best fit" will have a negative gradient.

As well as talking about the **form** (linear or non linear) of the relationship between two variables and the **direction** of the relationship (positive or negative) we can also talk about its **strength** (strong, moderate or weak).

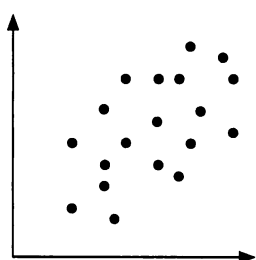
Positive and negative linear relationships of various strengths are shown below.



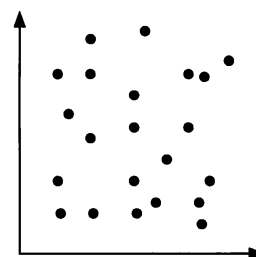
Perfect positive linear correlation.



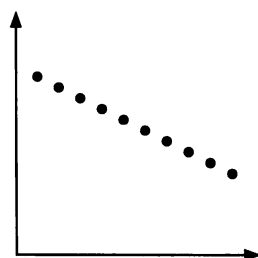
Strong positive linear correlation.



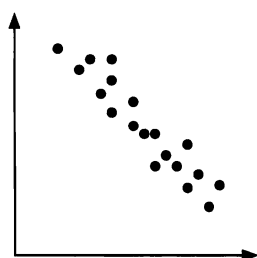
Moderate positive linear correlation.



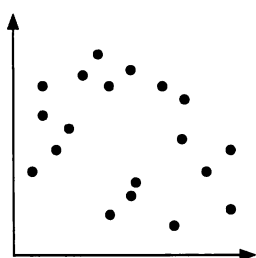
No linear correlation.



Perfect negative linear correlation.



Strong negative linear correlation.



Weak negative linear correlation.

Can we do better than “eye-balling” the line of best fit?

In chapter one we drew lines of best fit by “eye-balling” the scattergraph and, if a linear model seemed appropriate, drawing in the straight line that seemed to best fit the data. The weaknesses in this process is drawing the line of best fit “by eye”. What one person considers the best line may be quite different to the line chosen by someone else. Indeed it is easy to find yourself questioning your own choice of line moments after drawing it!

Thankfully many calculators can determine the line considered to be the best straight line, draw it, give its equation and use it to predict values.

Returning to the physics and maths test results of the previous page:

Physics Score	6	9	11	12	14	14	16	16	17	19
Maths Score	19	12	27	36	42	34	32	27	44	41

We would first enter these values into a calculator - a typical display is shown on the next page:

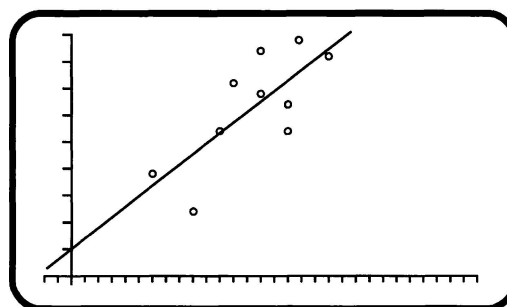
L1	L2	L3
6	19	
9	12	
11	27	
12	36	
14	42	
14	34	
16	32	
L2(1)=19		

The calculator can then use **linear regression** to:

- ➡ Output the equation of the line of best fit in the form, $y = mx + c$ where y is the maths score and x is the physics score (see display below left).
- ➡ Display the points and the line of best fit (see display below right).

LinearReg
 $y = ax + b$
 $a = 1.982905983$
 $b = 4.829059829$
 $r^2 = 0.5748032337$
 $r = 0.7581577894$

Line of best fit
 $y = 1.98x + 4.83$



- ➡ Use this line of best fit to predict a score in the maths test based on a score in the physics test. As shown on the right, for a physics score of 13.

Remember though that if **extrapolation** is involved (rather than **interpolation** as used here) any prediction would have to be viewed with caution.

y1(13) 30.60683761

- Note • The calculator display above left also shows values it calls r and r^2 . We will give these quantities further consideration later in this chapter. For now though let us concentrate our attention on the line of best fit and using it to make predictions.



Make sure that **you** can use **your** calculator to determine the line of best fit and the predicted value for the above bivariate data.

Can you get your calculator to display the scattergraph complete with the line of best fit?



Note • To make a distinction between the observed y values in our original data and the predicted y values, obtained from our regression line, the latter are sometimes written as $\hat{y}$, pronounced "y hat".

Using this notation for the predicted value of y on the previous page:

$$\text{when } x = 13, \quad \hat{y} = 30.6.$$

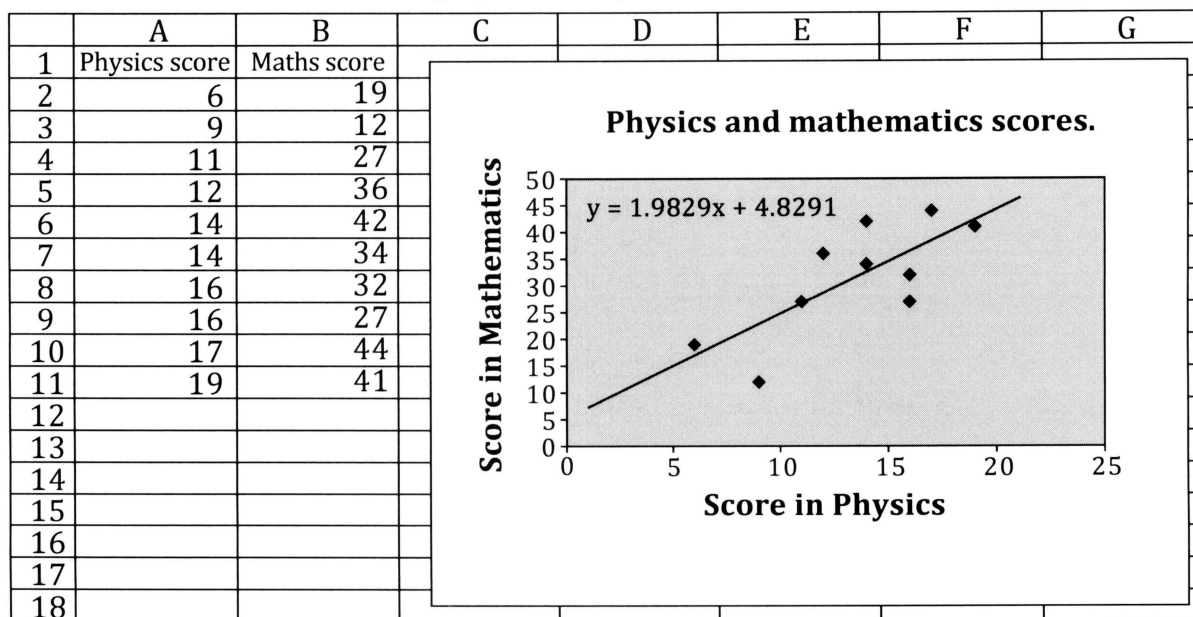
Similarly this "hat notation" can be used when stating the equation of the regression line:

$$\hat{y} = 1.98x + 4.83$$

This emphasises that the equation gives a predicted y value ($\hat{y}$) from a known x value.

Spreadsheets

Spreadsheets can also be used to display and analyse bivariate data in this way:



Try to use a computer spreadsheet yourself to display such data graphically.

Exercise 2A

- An investigation was carried out to see how the time (t seconds) a child took to complete a particular task depended on the age (y years) of the child ($5 \leq y \leq 18$). The data collected suggested that the situation could be well modelled using a line of best fit with equation

$$t = -1.5y + 41$$

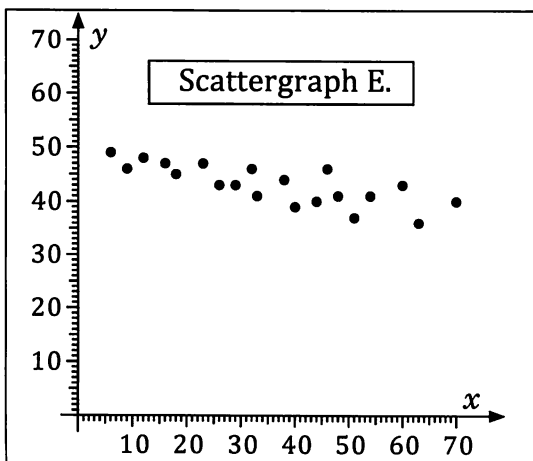
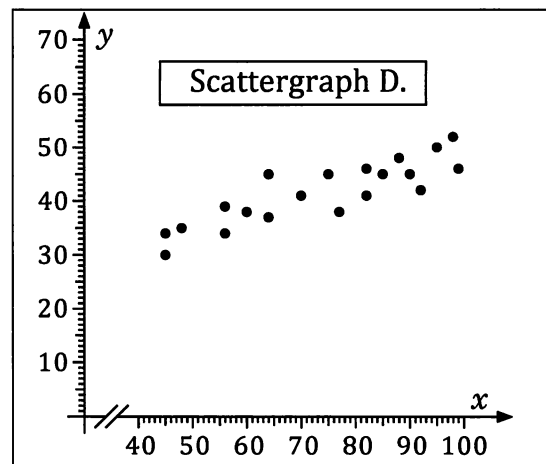
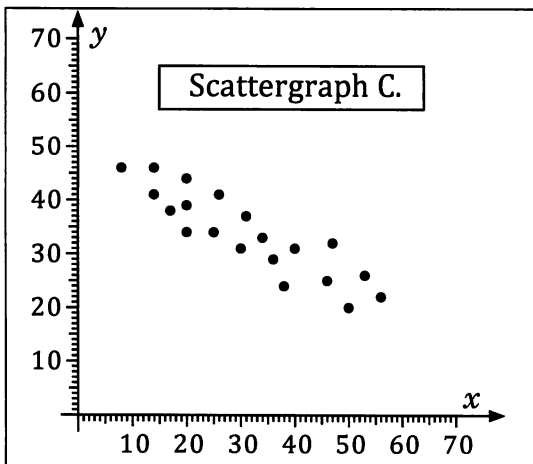
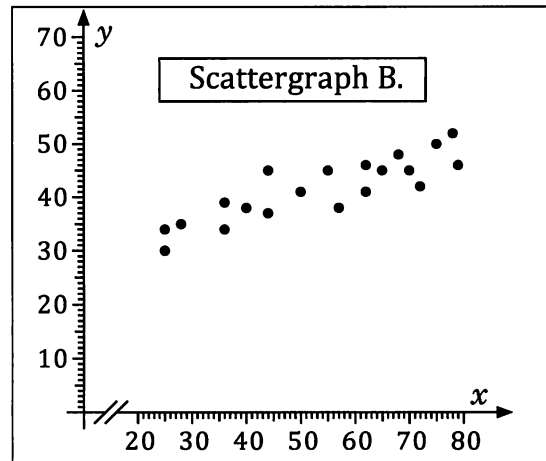
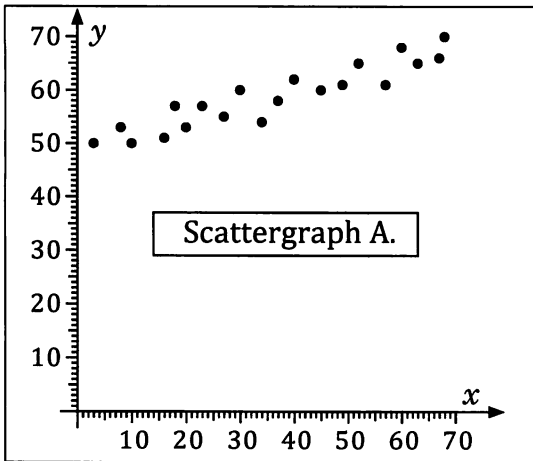
Interpret the numbers -1.5 and 41 in this situation.

- A study investigated the relationship between the time, h hours, that a student revises for an exam, and the mark, $m\%$, gained in the exam. The results suggested that the situation could be reasonably well modelled using the rule:

$$m = 2h + 28.$$

Interpret the numbers 2 and 28 in this situation.

3. The equations of the lines of best fit for the five scattergraphs A to E, shown below, are in the equations list also shown below.
For each scattergraph match the appropriate equation of the line of best fit. (You should be able to do this "by inspection", not by obtaining the line of best fit equation from your calculator.)



Equations.

$$y = -0.490x + 49.0$$

$$y = -0.152x + 48.6$$

$$y = 0.271x + 48.9$$

$$y = 0.283x + 26.4$$

$$y = 0.283x + 20.7$$

4. Seventeen students following a particular course had to take two tests, the first out of 40 and the second out of 25. One of the students was in hospital when the second test occurred and could not sit the test.

The marks obtained by the students are given below.

1st test	4	11	12	12	16	20	21	21	27	26
2nd test	11	9	15	4	16	13	16	20	18	Abs

1st test	28	30	30	35	35	39	39
2nd test	22	17	20	22	25	19	20

Draw a scatter diagram showing the results of the sixteen students who completed both tests.

Draw the line of best fit "by eye" and hence estimate a second test mark for the student who was unable to sit that test.

Use a spreadsheet or your calculator to create a scattergraph of the data complete with the line of best fit and compare it to your scattergraph and your line of best fit.

Determine the equation of the line of best fit from your calculator or spreadsheet.

Determine an estimated second test score for the student who was unable to sit that test using the calculator or spreadsheet's line of best fit and compare it to your "by eye" prediction.

5. Let us suppose that a medical check-up involves, amongst other things, measuring something called the *peak expiratory flow rate*, to detect possible respiratory problems. The peak flow rates achieved by twenty males, all aged between 30 and 50, are given in the table below, together with the height of each of these males.

Height (cm), h.	184	189	171	166	200	192	176
Peak flow rate (units/min), P.	500	555	510	550	625	610	525

Height (cm), h.	162	188	164	180	168	185	176
Peak flow rate (units/min), P.	500	570	475	530	520	575	570

Height (cm), h.	189	194	195	190	191	172
Peak flow rate (units/min), P.	525	600	560	590	585	540

Draw a scattergraph of these figures with the height, h, on the horizontal axis and peak flow, P, on the vertical axis and draw the line of best fit "by eye".

Hence estimate the peak flow rate for a male aged between 30 and 50 and of height 174 cm.

Now use a calculator or spreadsheet to display the data as a scattergraph, obtain the equation of the line of best fit and the predicted peak flow rate for a male aged between 30 and 50 and of height 174 cm and compare the value with your "by eye value".

6. Sixteen students following a particular course had to take two examinations, one in mathematics and one in physics. One of the students was taken ill during the physics exam and could not complete the exam.

The marks obtained by the students are given below.

Initials	JA	FC	ED	SF	JG	KK	ML	TR
Maths (%)	13	87	55	30	63	11	68	40
Physics (%)	30	49	35	41	ill	20	60	34

Initials	RR	RS	JS	WT	AV	ZW	LW	PY
Maths (%)	85	30	58	72	40	20	95	45
Physics (%)	56	25	43	52	55	27	67	54

Either by drawing it, or by using a calculator or spreadsheet, view a scattergraph showing the results of the fifteen students who completed both tests with M , the maths percentage on the horizontal axis, and P , the physics percentage, on the vertical axis.

Comment on the suitability of using a linear model.

Determine the equation of the line of best fit in the form $P = aM + b$ and use this to determine an estimated physics mark for the student who was unable to complete that exam.

7. The table below gives the *Infant Mortality Rate* (deaths per 1000 infants) and the *Literacy Rate* (percentage of population who are literate) for 15 countries.

Country	A	B	C	D	E	F	G	H
Literacy Rate (%), L .	64	81	52	87	78	95	35	99
Infant Mortality, I .	43	60	78	27	52	29	107	9

Country	I	J	K	L	M	N	O
Literacy Rate (%), L .	93	77	52	98	35	69	42
Infant Mortality, I .	15	67	76	17	102	74	90

- With the literacy rate on the horizontal axis, and the mortality rate on the vertical axis, view the data as a scattergraph either by drawing it or by displaying it on a calculator or spreadsheet.
- Determine the equation of the line of best fit in the form $I = aL + b$.
- Hence determine an estimate of the infant mortality rate for a country
 - with a literacy rate of 48%,
 - with a literacy rate of 15%,
 and in each case comment on how reliable your predictions might be.

Pearson's correlation coefficient.

Describing a relationship between two variables using terms such as *strong positive* or *weak negative* is rather imprecise. In this section we will see how the strength of a relationship can be stated in terms of a **correlation coefficient**, a number from -1 to 1 (also known as Pearson's correlation coefficient).

In the calculator display shown on an earlier page, and shown again on the right, "r" is the correlation coefficient.

The display also gives r^2 , the square of the correlation coefficient, called the *coefficient of determination*. We will consider this later but for now let us concentrate our attention on the correlation coefficient, r .

LinearReg

$y=ax+b$

$a=1.982905983$

$b=4.829059829$

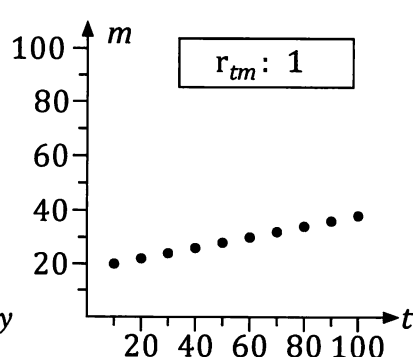
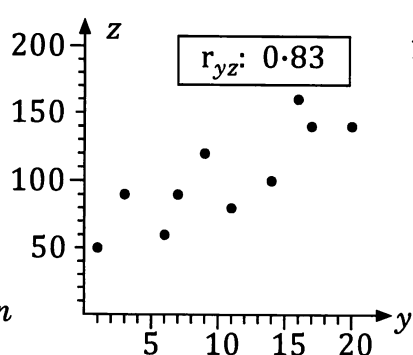
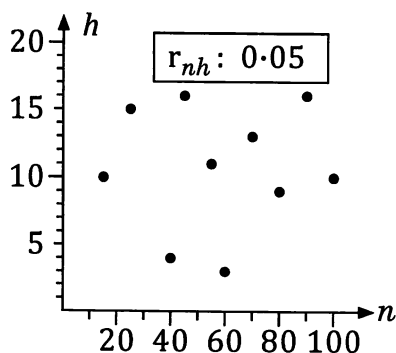
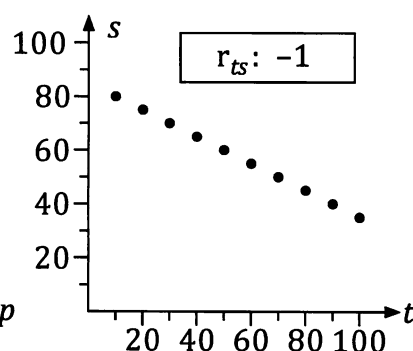
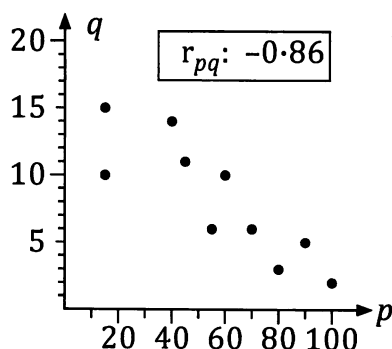
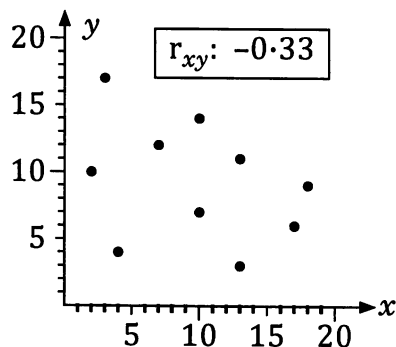
$r^2=0.5748032337$

$r=0.7581577894$

The **correlation coefficient** is a measure of how closely the relationship between two sets of data approximates to a linear relationship. We tend to use r for the correlation coefficient where $-1 \leq r \leq +1$. We sometimes write the correlation coefficient as r_{xy} where the subscripted letters are the two variables involved, in this case x and y .

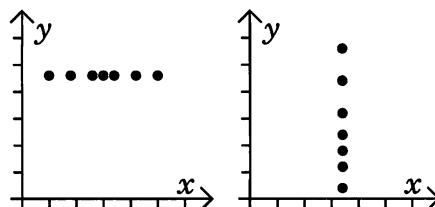
A correlation coefficient of -1 indicates a perfect linear relationship with a negative gradient (as one variable increases the other decreases) and a correlation coefficient of $+1$ indicates a perfect linear relationship with a positive gradient (as one variable increases the other increases).

A few scattergraphs and the respective correlation coefficients are shown below.



Note • Questions in this text may deliberately involve only about ten data points or so to allow you to practice the techniques involved without spending too much time processing the data. In reality, to make more valid any comments as to whether two variables show correlation, we would try to involve more data points.

- The graphs on the right both show points that lie in a perfect straight line but in both cases the idea of a correlation coefficient is meaningless because we do not have *bivariate* data. In each case either x or y is not showing any variation at all.



Using your calculator to determine r .

As we have already seen, the value of r and the equation of the line of best fit can be determined using a calculator, or computer, that can perform **linear regression**, the name given to the process of determining the best straight line for bivariate data. (The line of best fit is sometimes referred to as the **regression line**.)

For the following bivariate data the value of r and the equation of the linear regression line are as given below the table.

x	92	23	68	64	57	81	12	98	32	20	47	96	81	40
y	51	32	39	34	25	36	21	36	21	20	35	43	42	26

Correlation coefficient, r_{xy} 0.83 (correct to 2 d.p.)

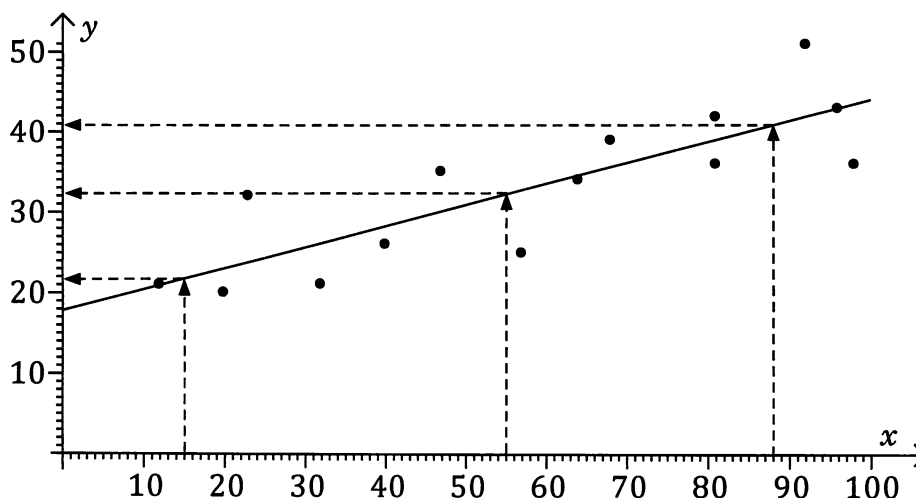
Line of best fit is $y = ax + b$ where $a = 0.262$ and $b = 17.76$.

For $x = 15$ the line of regression gives a predicted y value of 21.7.

For $x = 55$ the line of regression gives a predicted y value of 32.2.

For $x = 88$ the line of regression gives a predicted y value of 40.8.

The graph below shows the bivariate data, the regression line and the predicted values.



Can we have confidence in our predictions?

Notice that all three of the predicted values on the previous page involve **interpolation** - a term introduced in the previous chapter.

Hence with

- ① the distribution of the data points suggesting that a linear model would be appropriate,
 - ② the value for r of 0.83 confirming the presence of a reasonably strong positive linear correlation,
 - ③ a reasonable number of points being involved,
- and ④ the fact that interpolation is involved (*between* known data points), rather than extrapolation (*beyond* the known data points)

we can be reasonably confident of the validity of our predicted values.

Example 1

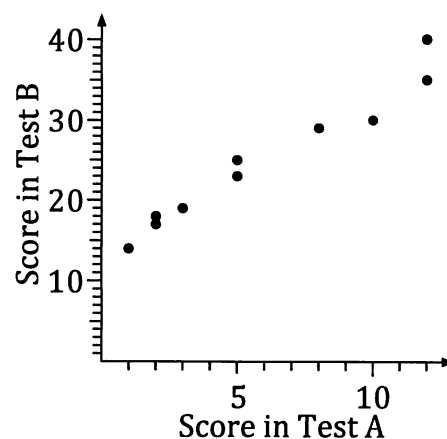
Ten people were given two tests, A and B and the marks obtained were as follows:

Test A	1	2	2	3	5	5	8	10	12	12
Test B	14	17	18	19	23	25	29	30	35	40

- (a) Plot the data as a scattergraph with the test A results on the horizontal axis and the test B results on the vertical axis.
- (b) Suggest whether or not the data seems suited to the use of a linear model.
- (c) Determine the correlation coefficient for the 10 pairs of results.
- (d) Determine the equation of the line of best fit in the form

$$\text{Test B score} = a \times \text{Test A score} + b$$
- (e) Predict the Test B score for someone with Test A score of 7 and comment on the validity of your prediction.

- (a) The required scattergraph is shown on the right.
- (b) The points on the scattergraph suggest a strong positive linear correlation exists between the scores on test A and the scores on test B. Hence the data seems well suited to the use of a linear model.
- (c) Putting the given data pairs into a calculator we obtain $r = 0.978$ (3 d.p.).
 (Further confirming the suitability of a linear model.)



- (d) The line of best fit has equation: Test B score = $1.956 \times \text{Test A score} + 13.26$
- (e) For a test A score of 7, predicted test B score = $1.956 \times 7 + 13.26$
 ≈ 27

With a reasonable number of points involved, the high value of r and interpolation involved we can be reasonably confident that our predicted value is valid.

Cause and effect.

Though a scattergraph, and an r value, may suggest that a correlation exists between two variables we cannot assume that the changes in one variable *cause* the changes in the other variable.

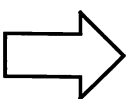
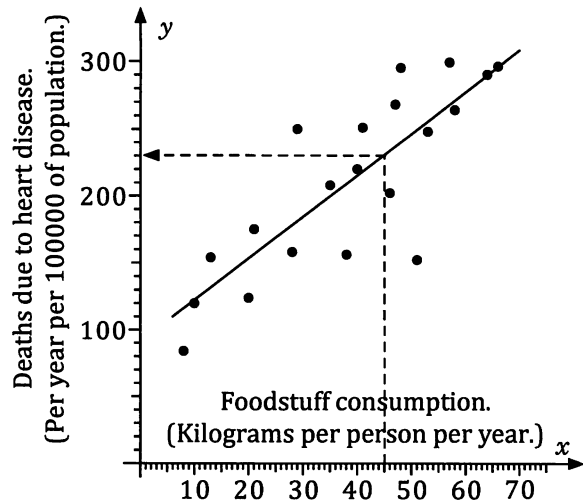
In an earlier example we found there to be a correlation, or association, between the consumption of a particular food stuff and the death rate due to heart disease. However this does not prove that the foodstuff consumption *causes* the increased death rate. The graph suggested a relationship, not a "cause and effect".

There could be other factors at work. For example it could be that the high intake of the foodstuff indicates a more affluent society and it is the lifestyle of the more affluent society that is the reason for the higher death rate due to heart disease.

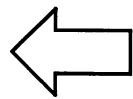
This *affluence of society* is a **lurking variable** or **confounding variable** that each of our two variables, food consumption and heart disease death rates, could have a **common response** to.

To take another example, we would probably not be surprised if the monthly sales of ice-cream and the monthly number of deaths due to drowning were correlated. However in this case the lurking variable, or confounding variable, is likely to be *temperature*. On hot days more ice-creams are sold and more people go swimming, hence the correlation between ice cream sales and deaths due to drowning. It would be incorrect to conclude that because there is correlation, one is causing the other.

It is also possible that collected data could suggest an association between two variables but the apparent association is purely **coincidental**. For example a person might find that on several occasions, when they buy some seedlings for planting in the garden, the weather for the next few days after planting seems to be very hot, causing the seedlings to die. However this is likely to be just coincidence and certainly not some causal relationship between the seedlings being planted and a spell of hot weather following. A company making cars might find that sales of their cars have been dropping ever since a particular employee started working for the company, but again this is likely to be coincidence, not any causal relationship between the sales figures and the length of time the employee has been with the company.



Hence we need to recognize that an observed association or correlation between two variables does not necessarily mean that there is a causal relationship between the variables. There might be a causal relationship but we cannot assume it exists purely on the basis of there being an association.



Strong, moderate or weak relationship?

For what values of r do we describe the linear relationship between two variables as being strong, moderate or weak?

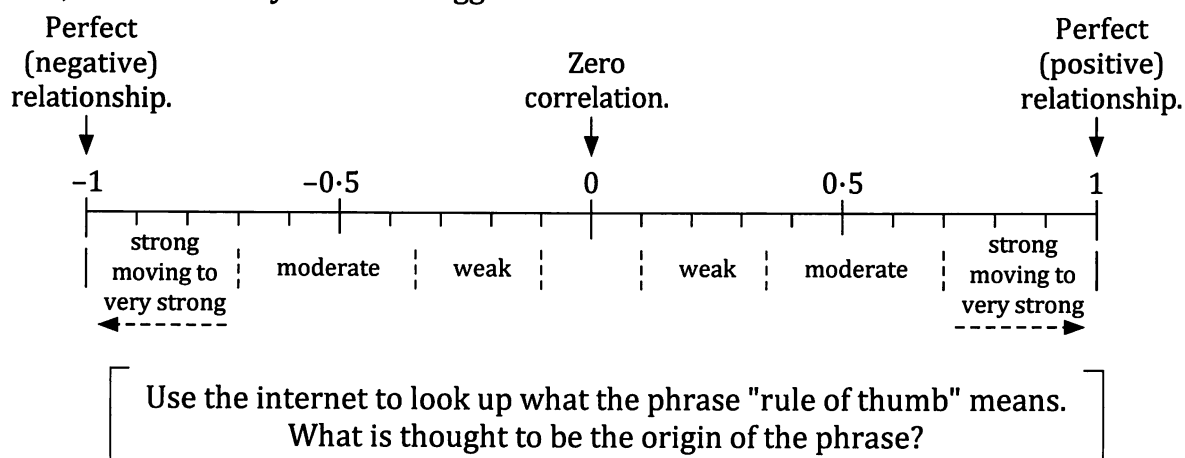
It is not quite as clear cut as setting rules like "for r between 0.3 and 0.5 we say that the linear correlation is weak positive". Other aspects may need consideration.

How many data points were used to determine r ?

Are "outliers" lowering the value of r ?

Some environments, for example medical, may require a higher value of r before wanting to suggest a "strong" relationship exists.

To attempt to give some idea of when terminology such as strong, moderate or weak might be considered reasonable, for various r values, the diagram below gives a "rule of thumb" that readers might consider useful. However, as an internet search would reveal, there are many different suggestions for where the boundaries should lie.

**About the correlation coefficient, r_{xy} .**

Whilst you should understand what the correlation coefficient tells you about bivariate data you would not normally be expected to calculate its value yourself, preferring instead to obtain its value using the statistical capability of a calculator. However you should be able to read and understand the next few paragraphs to get some understanding of how the correlation coefficient is calculated.

The scattergraph shown on the right shows the bivariate data of example 1, with test A scores shown on an x -axis and test B scores on a y -axis.

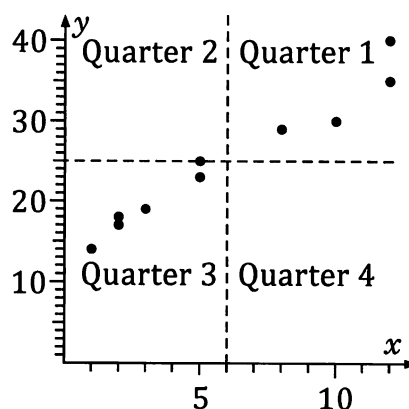
Dashed lines are drawn at the mean of the x scores, $\bar{x}$, and the mean of the y scores, $\bar{y}$.

These dashed lines divide the graph into 4 quarters.

For a strong positive linear relationship we would expect most points to lie in quarters 1 and 3, which we can see is indeed the case for this scattergraph.

For points in these quarters the quantities $(x \text{ score} - \bar{x})$ and $(y \text{ score} - \bar{y})$ are either both positive or both negative.

Thus the mean of all the products $(x \text{ score} - \bar{x})(y \text{ score} - \bar{y})$ will be positive.



For bivariate data with a strong negative linear relationship many of the points will lie in quarters 2 and 4 where the quantities $(x \text{ score} - \bar{x})$ and $(y \text{ score} - \bar{y})$ are of opposite sign. The mean of all the products $(x \text{ score} - \bar{x})(y \text{ score} - \bar{y})$ will then be negative.

Thus this process gives a number that is positive for bivariate data showing a positive linear relationship and negative for data showing a negative linear relationship.

We call this mean of the products $(x \text{ score} - \bar{x})(y \text{ score} - \bar{y})$ the **covariance** of x and y , written s_{xy} :

Hence if we have n data points $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots (x_n, y_n)$.

$$\text{Covariance, } s_{xy}, = \frac{(x_1 - \bar{x})(y_1 - \bar{y}) + (x_2 - \bar{x})(y_2 - \bar{y}) + \dots + (x_n - \bar{x})(y_n - \bar{y})}{n}$$

For the ten pairs of scores on the scattergraph on the previous page $\bar{x} = 6$ and $\bar{y} = 25$.
For these points:

x	1	2	2	3	5	5	8	10	12	12	
y	14	17	18	19	23	25	29	30	35	40	
$x - \bar{x}$	-5	-4	-4	-3	-1	-1	2	4	6	6	
$y - \bar{y}$	-11	-8	-7	-6	-2	0	4	5	10	15	Total
$(x - \bar{x})(y - \bar{y})$	55	32	28	18	2	0	8	20	60	90	313

$$\begin{aligned}\text{Covariance, } s_{xy}, &= \frac{313}{10} \\ &= 31.3\end{aligned}$$

However the magnitude of this covariance will depend on the magnitude of the original x and y values. To overcome this we then divide by the standard deviations of the x and y sets of data. (The size of these quantities are similarly dependent upon the size of the original data and so division will cancel this effect.) This will give a number that is positive for bivariate data showing a positive linear relationship, negative for data showing a negative linear relationship, and will be independent of the size of the original data. This number is the **correlation coefficient**, r_{xy} , and is such that $-1 \leq r \leq 1$.

$$r_{xy} = \frac{\text{Mean of the products: } (x \text{ score} - \bar{x})(y \text{ score} - \bar{y})}{(\text{Standard deviation of the } x \text{ scores})(\text{Standard deviation of the } y \text{ scores})}$$

$$\begin{aligned}\text{i.e. } r_{xy} &= \frac{s_{xy}}{s_x s_y} \\ &= \frac{31.3}{(4)(8)} \quad (\text{for the data given above}) \\ &= 0.978125\end{aligned}$$

Confirm that this is the value your calculator gives for the correlation coefficient for this set of ten data pairs.

About the line of best fit.

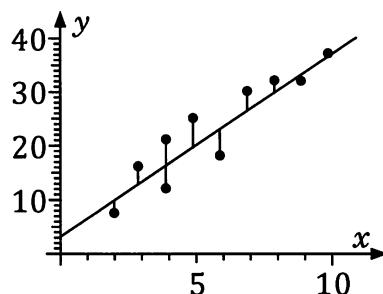
The scattergraph on the right involves ten points and as well as showing the points, and their line of best fit, it also shows the vertical deviations of each point from the line of best fit.

These lines show the differences between the obtained y values and the y values predicted by the regression line.

These differences are called the linear regression **residuals**.

i.e. Residual = Observed y value – y value predicted by the regression line.

$$= y - \hat{y}$$



When a calculator determines the line of best fit for estimating y for given values of x it is finding the line which minimises the *sum of the squares of these vertical deviations*.

We call this the line of best fit, the trend line, or the

least squares regression line of y on x .

It is used to predict **y based on x** .

- Note
- The fact that our regression line is determined by minimising the sum of the *squares* of the deviations of each point from the line means that any outlying points can greatly influence the location of the regression line.
 - Prior to the availability of calculators capable of processing statistical data the least squares regression line was determined using the rule:

Regression line, y on x , is
$$y - \bar{y} = \frac{s_{xy}}{s_x^2} (x - \bar{x}) .$$

- Notice that if we substitute $x = \bar{x}$ into the regression line equation from the previous dot point:

$$y - \bar{y} = \frac{s_{xy}}{s_x^2} (\bar{x} - \bar{x})$$

i.e.
$$y - \bar{y} = 0$$

and so
$$y = \bar{y}$$

Thus:

The least squares regression line passes through $(\bar{x}, \bar{y})$.

- We predict y using the **y on x regression line**, a line determined by minimising the sum of the squares of the vertical deviations of each point from the line. We should **not** use the equation of this y on x line "in reverse" to estimate an x value given a y value. Instead we use the **x on y regression line**, a line determined by minimising the sum of the squares of the horizontal deviations.

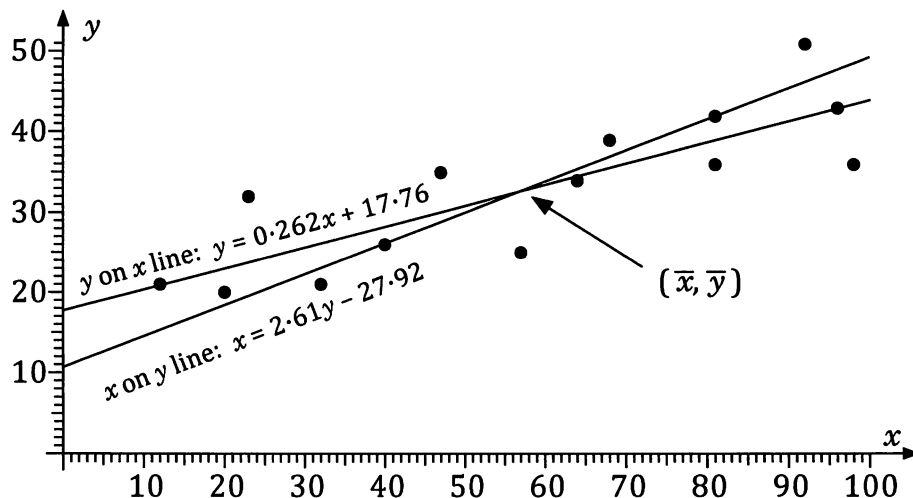
Hence the **y on x regression line** is used to predict y **based on** x and the **x on y regression line** is used to predict x **based on** y .

With the explanatory variable as our x values and the response variable as our y values we would usually want to predict y , given x . Hence it will usually be the y on x regression line that we will use.

For r close to ± 1 there will be little difference between these lines and they will coincide if $r = \pm 1$.

The graph shown below shows both regression lines for the following data, encountered earlier in this chapter:

x	92	23	68	64	57	81	12	98	32	20	47	96	81	40
y	51	32	39	34	25	36	21	36	21	20	35	43	42	26



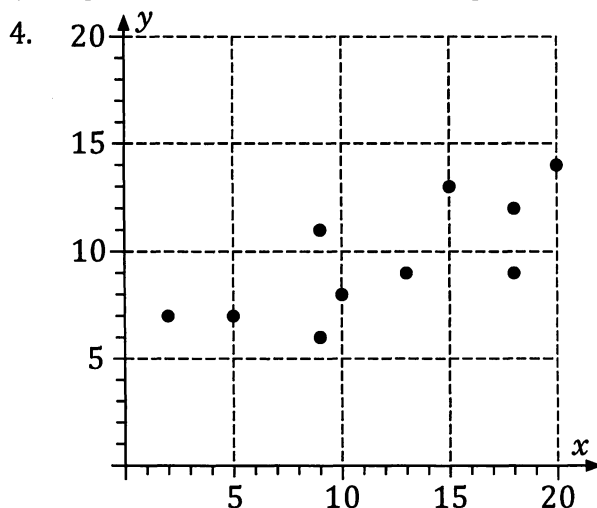
Make sure you can obtain the equation of the x on y regression line.
(Instruct your calculator to read the ordered pairs as (y, x) when it calculates the line of best fit).

Exercise 2B

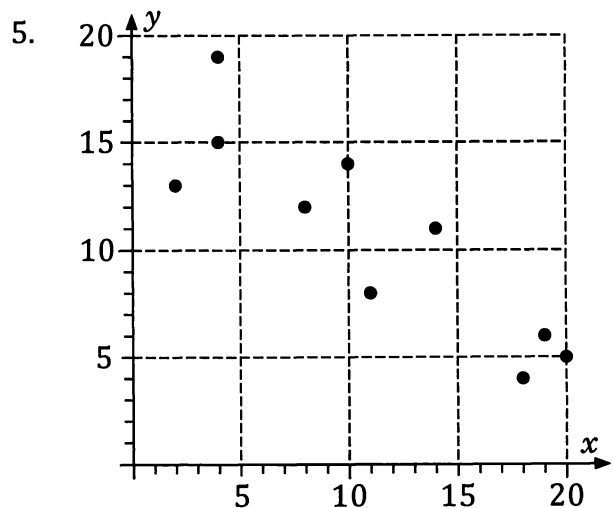
- Suppose that a survey of teenagers found that there was an association between the number of hours spent reading teen fashion magazines and the chance of the teenager having an eating disorder.
Can we conclude that the reading of fashion magazines is causing the eating disorder?
Write a few sentences explaining what this association might be due to.
- From the time that a particular individual started his employment as head gardener at the XYZ export factory the company profits have been in steady decline. As the length of time that the individual has been with the company has increased then so the profits have decreased.
Can we conclude from this association between length of employment of the individual and company profits that it is the individual who is causing the decline in company profits?
Write a few sentences discussing this situation.
- A survey found that over recent years the average age of school leavers had shown a steady increase and so too had the number of school leavers having attention deficit disorder.
Could we conclude from this association between years at school and cases of attention deficit disorder that it is the extra time at school that is causing the attention deficit disorder?
Write a few sentences discussing this situation.

For the data given in each of questions 4 to 11 use your calculator to determine

- the correlation coefficient r_{xy} ,
 - the equation of the regression line of y on x : $y = Ax + B$,
 - the predicted y values requested in each question and state for each predicted value whether it is *interpolation* or *extrapolation* that is involved.
- (For questions 4 to 7 assume integer coordinates.)

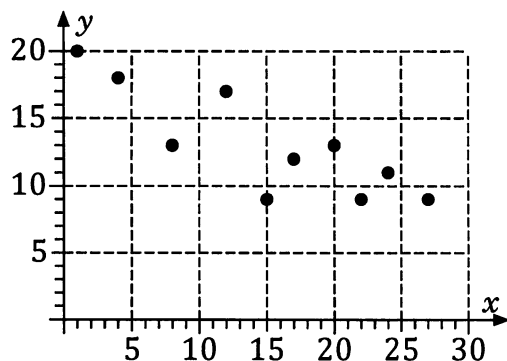


Predict the y value for $x = 28$.



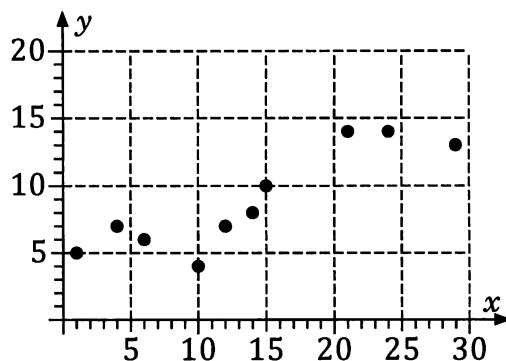
Predict the y value for $x = 6$.

6.



Predict the y value for (i) $x = 10$,
(ii) $x = 40$.

7.



Predict the y value for (i) $x = 40$,
(ii) $x = 17$.

8.

x scores	y scores
1	2.7
2	3.1
3	2.5
4	2.5
5	3.1
6	3.4
7	3.2
8	3.7
9	3.1
10	3.5

Predict the y value for $x = 11$.

9.

x scores	y scores
18	43
40	30
25	28
4	50
54	15
45	15
59	5
33	14
54	10
10	30

Predict the y value for (i) $x = 22$
(ii) $x = 47$

10.

x scores	y scores
4.5	340
1.7	320
4.1	360
5.7	500
2.5	280
7.2	520
3.7	410
6.5	530
3.2	410
1.3	280
0.8	300
5.6	390

Predict the y value for (i) $x = 5.1$,
(ii) $x = 9.2$,
(iii) $x = 12.9$.

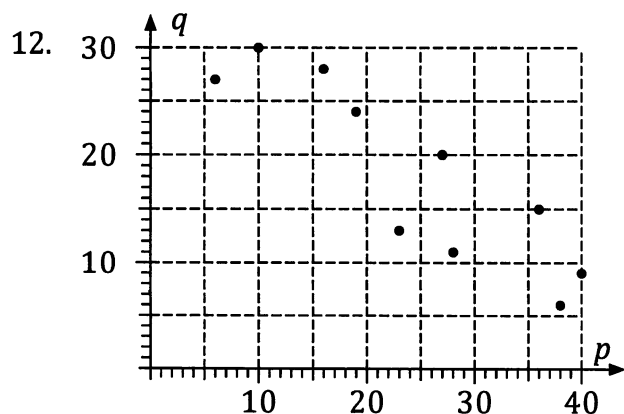
11.

x scores	y scores
95	65
50	29
20	5
120	85
35	17
65	41
110	77
70	45
45	25
30	13
60	37
105	73

Predict the y value for (i) $x = 85$,
(ii) $x = 40$,
(iii) $x = 200$.

For the data given in each of questions 12 and 13 below determine

- the equation of the regression line of q on p : $q = ap + b$.
 - the equation of the regression line of p on q : $p = cq + d$.
 - the predicted values requested using the appropriate regression line.
- (For question 12 assume integer coordinates.)



- Predict (i) the q value for $p = 30$,
(ii) the p value for $q = 25$.

13.

p scores	q scores
35	19
20	16
60	24
45	21
5	13
95	31
25	17
85	29
55	23

- Predict (i) the q value for $p = 48$,
(ii) the p value for $q = 32$.

14. The owner of a fish and chip shop feels sure that on any summer's evening the number of customers buying fish and chips is dependent upon the temperature that evening. To test this theory one summer he notes the temperature at 5 pm when he opens his shop and also notes the number of serves of fish he sells that evening.

**It's
Summertime**

A period of 15 summer evenings produced the following results:

Evening	1	2	3	4	5	6	7	8
Temp ($T^{\circ}\text{C}$)	31	35	24	33	34	25	37	33
N ^o . of serves (N)	202	83	208	141	128	157	72	112

Day	9	10	11	12	13	14	15
Temp ($T^{\circ}\text{C}$)	30	27	31	29	34	33	28
N ^o . of serves (N)	142	203	104	170	99	157	121

- Determine the correlation coefficient r_{TN} .
- Determine the equation of the line of regression $N = A + BT$.
- Predict the value of N for (i) $T = 26$ and (ii) $T = 42$ commenting on the likely validity of your predictions in each case.

15. The shop keeper of question 14 feels that during winter the trend is different. The lower temperatures now make going out to get dinner rather unpleasant and people are less likely to visit his shop on very cold evenings, preferring instead to use a food outlet that has a delivery service (which his does not). Again he notes the temperature at 5 pm when he opens his shop and the number of serves of fish he sells that evening. A period of 15 winter evenings produced the following results:



Evening	1	2	3	4	5	6	7	8
Temp ($T^{\circ}\text{C}$)	2	16	15	8	6	19	4	9
N ^o . of serves (N)	40	153	192	104	98	180	52	130

Day	9	10	11	12	13	14	15
Temp ($T^{\circ}\text{C}$)	10	3	13	17	7	11	6
N ^o . of serves (N)	173	91	230	242	51	130	131

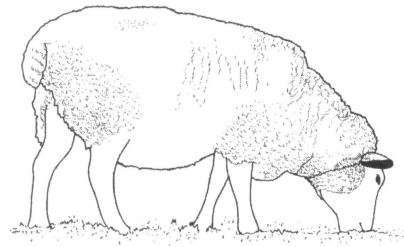
- (a) Determine the correlation coefficient r_{TN} .
- (b) Determine the equation of the line of regression $N = A + BT$.
- (c) Predict the value of N for (i) $T = 14$ and (ii) $T = -3$ commenting on the likely validity of your predictions in each case.
16. In a particular course sixteen students sat a mid-year exam, marked out of 80, and an end of year exam, marked out of 120. The results they obtained were as follows:

Initials	TB	KD	MF	KH	TH	LK	PL	RL
Mid Year (out of 80)	47	72	54	32	78	53	65	74
End Year (out of 120)	82	98	92	56	105	81	79	115

Initials	SM	TN	WN	KS	RS	TT	VW	PY
Mid Year (out of 80)	52	53	42	67	60	67	43	66
End Year (out of 120)	71	84	73	104	104	88	54	100

- (a) Determine the correlation coefficient for the above set of results.
- (b) Determine the equation of the least squares regression line for predicting end of year marks based on mid year marks.
- (c) Another student scored 63 out of 80 in the mid year exam but did not sit the end of year exam due to a family bereavement. Use your regression line to estimate an end of year exam mark for this student.

17. Scientists are testing a new spray for its effectiveness in controlling a particular weed that is toxic to sheep. The scientists test various concentrations of the spray over test areas of 1 hectare in which the weed is prevalent. The scientists then determine the percentage of the weed in this 1 hectare test area that survives the spray one week after its application. The results obtained are given in the table below.



Concentration (C%)	10	25	30	45	50	55	60	70	75	80
Survival rate (S%)	32	23	24	20	18	12	15	9	8	2

- (a) Determine the correlation coefficient r_{CS} and the equation of the least squares regression line for determining S based on C .
- (b) It is likely that if the spray is allowed into widespread use the maximum concentration allowed will be 38%, for health reasons. If spray of this level of concentration is applied to an area in which the weed is prevalent estimate the percentage of the weed which would not survive the spray one week after its application?
18. The table below gives the life expectancy in years for ten countries and also the percentage of the population of those countries having access to good sewage disposal and toilet facilities (i.e. good sanitation facilities).

Country	A	B	C	D	E	F	G	H	I	J
Percentage with access to good sanitation ($S\%$)	26	29	31	43	53	75	87	92	94	97
Life Expectancy (L yrs)	59	61	52	62	63	71	67	68	76	70

- (a) Determine the correlation coefficient r_{SL} and the equation of the least squares regression line for determining L based on S .
- (b) Hence predict the life expectancy for a country with
- 60 % of its population with access to good sanitation facilities,
 - 80 % of its population with access to good sanitation facilities.
- (c) Can we conclude that a high life expectancy is due to having access to good sanitation?

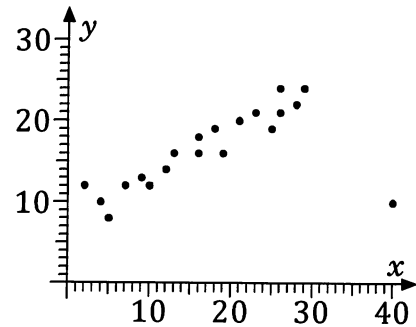
Some further considerations.

1. Outliers.

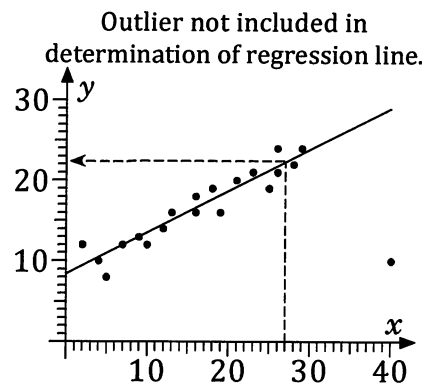
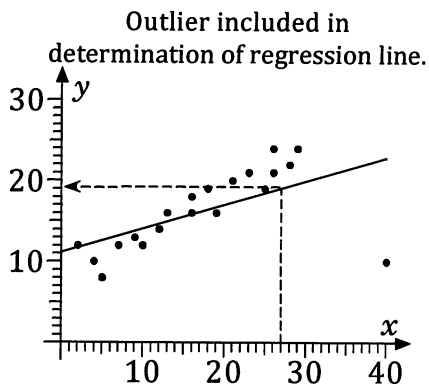
Consider the scattergraph shown on the right. The point (40, 10) seems to be very much out of step with the reasonably strong linear trend suggested by the other points.

The point (40, 10) is an **outlier**.

Outliers can significantly alter the location of the line of best fit. The two versions of the scattergraph shown below include the line of best fit with the outlier included in the determination of the line below left and with the outlier excluded below right.



Notice how the predicted y -value for an x -value of 27 differs in the two graphs.



Determining the line of best fit involves finding the straight line that best fits the data points and summarises the overall trend of the points. The presence of an outlier can greatly affect the calculation of the equation of this line. An outlier can have a greater effect on the determination of the line of best fit than a single point should have. To get a straight line that best summarises most of the points we may choose to leave the outlier out when determining the line of best fit. However it is important to realise that whilst in this section we are suggesting that an outlier should be removed to avoid it having an undue influence on the determination of the line of best fit, we are not suggesting that whenever there are outliers in statistical data, remove them. In some areas of investigation the one response that seems out of line with all of the others could be the very thing we are searching for. Consider for example a research worker testing many different drugs to see which, if any, have any effect on a certain disease. In this case the one drug that produces results that are different to all of the others could be the very one that holds the key to defeating the disease. The response should be to *investigate outliers further* rather than just reject them without thought. However with scattergraphs, if we are wanting to determine a line that best summarises the overall trend suggested by the majority of the points, we may need to remove outliers from the calculation to avoid them having too significant an effect on the line of best fit.

Do a search on the internet to investigate how the automatic rejection of outliers apparently delayed the detection of the thinning of the ozone layer.

2. "Trimming" or "cropping" bivariate data.

Given a set of data containing an outlier we may choose to remove the outlier when analysing the data. This could be viewed as *trimming* or *cropping* the data and "harvesting" only some of it for analysis.

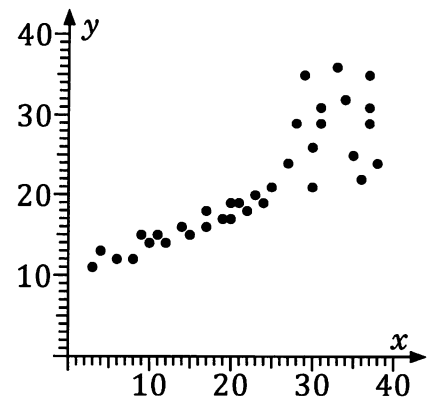
Consider the scattergraph on the right.

The strong linear trend suggested for the data points with the smaller x values seems not to apply at all for x values over 25.

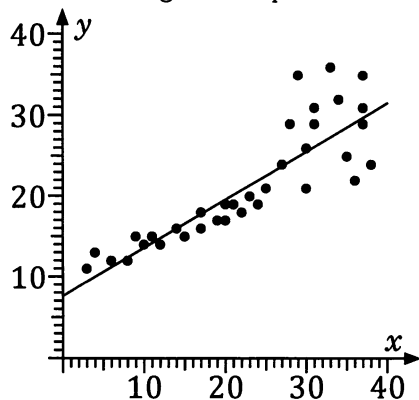
If we wanted to make predictions based on x values in the range 3 to 25 we could crop the data so that it only included points for which

$$3 \leq x \leq 25$$

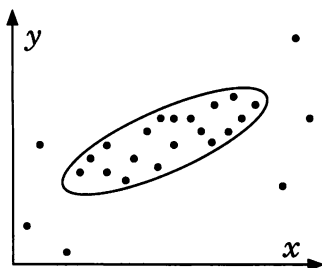
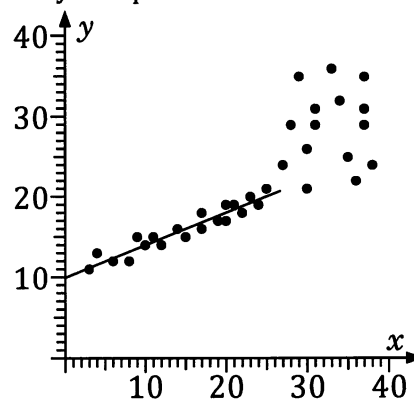
and determine and use the line of best fit for this cropped data only.



Regression line determined using all data points.



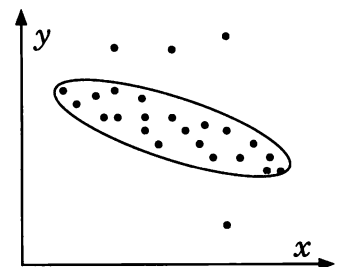
Regression line determined using only data points for which $3 \leq x \leq 25$.



The scattergraph shown on the left seems to have a number of outliers for the lower and higher values of x but the data points in the middle range of x values seem to have quite a strong linear correlation.

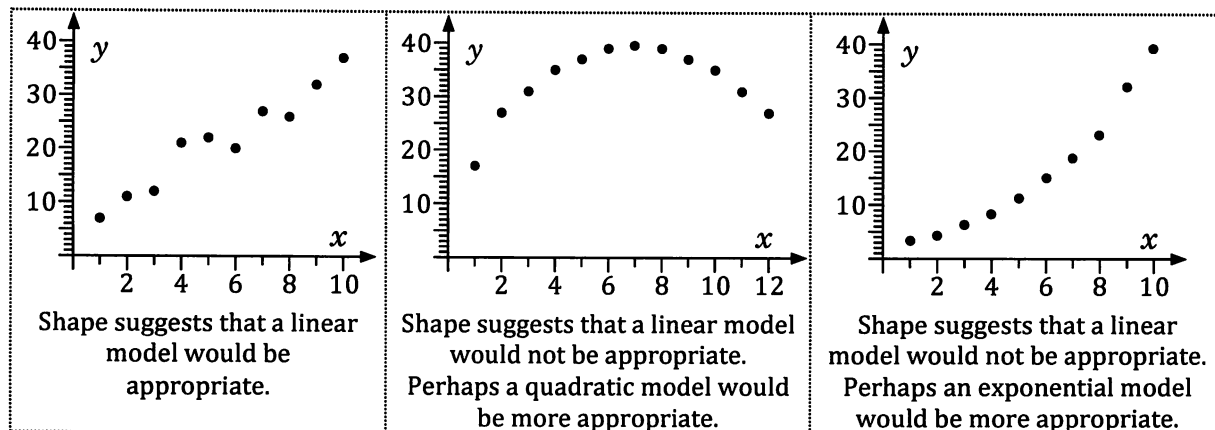
In this case we might consider cropping the data so that our line of best fit is calculated using only those data points lying in the oval shown on the graph.

Similarly, for the scattergraph shown on the right, we might consider rejecting the four outlying points and only consider the trimmed or cropped data lying in the oval shown.

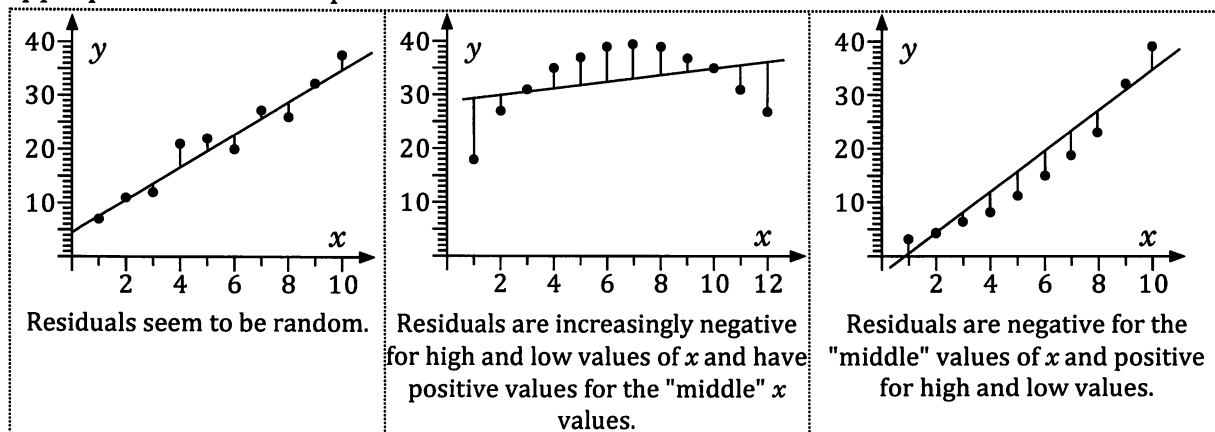


3. Residual plots.

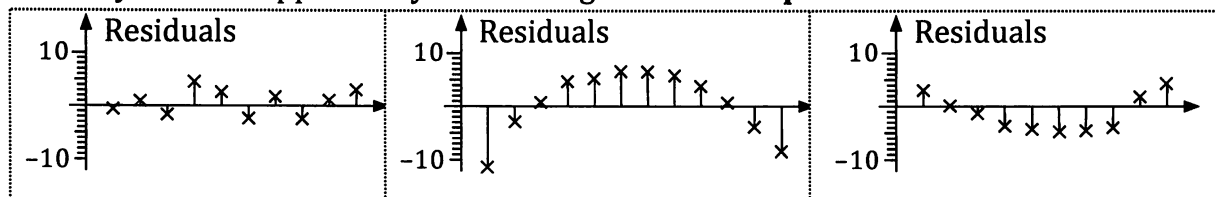
So far this chapter has concentrated on determining the *straight* line that best fits bivariate data. We have used *linear* regression techniques to model sets of ordered pairs (x, y) in an attempt to predict the likely value of one variable given the value of the other. However, will a straight line always be the best choice of regression model? Consider the three scattergraphs given below.



Notice also that if we were to attempt to fit a straight line to each of the above sets of bivariate data the **residuals** (observed y value minus $\hat{y}$, the value predicted by the regression line) are reasonably random for the data set for which linear regression is appropriate but form a pattern in the other two.



This may be more apparent by considering the **residual plots** shown below.



If we did not have the initial graphs to look at but did have a table of residuals, or a residual plot, then noticing the random nature of the residuals can be an indication that a linear model would be appropriate (or a non-random set of residuals indicates that a linear model may not be the most appropriate).

4. r^2 , the coefficient of determination.

A question encountered earlier in this book involved the advised cooking time for a joint of meat, according to the weight of the joint. The scattergraph for the situation is shown on the right. Notice that there is a perfect linear association between the cooking time and the weight of the joint. The correlation coefficient, r , is 1.

The variation in the cooking time is 100% explained, or accounted for, by the relationship between the two variables, cooking time and weight of joint. Knowing the weight of the joint we can exactly determine the advised cooking time.

Now let us again consider the food consumption / heart disease death rate situation encountered previously.

For this data

Least squares regression line:

$$y = 3.1x + 90.9$$

Correlation coefficient, r (to 3 dp):

$$0.823$$

Coefficient of determination, r^2 (to 3 dp):

$$0.677$$

The variation in death rate due to heart disease is not 100% explained by the variation in the foodstuff consumption.

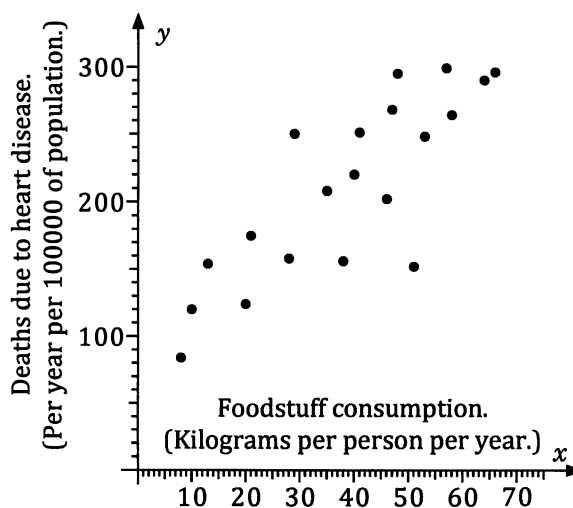
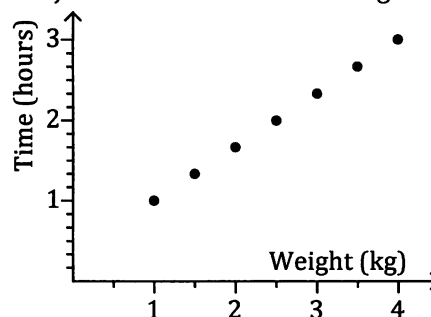
However we can use r^2 to state what proportion of the total variation in the death rate due to heart disease is explained, or accounted for, by the variation in the foodstuff consumption. With an r^2 value of 0.677 we can say that approximately 68% of the variation in death rates due to heart disease can be explained, or accounted for, by the variation in the foodstuff consumption.

Note carefully that we are saying "explained by", not "caused by". There is an association between the two variables, so as one changes we should expect the other to change, but that is not to say that the changes in one variable cause the changes in the other.

If a linear association exists between two variables such that the correlation coefficient is r , then r^2 , the coefficient of determination, is the proportion of the variation that can be explained by the linear relationship.

For example suppose that a survey found that the correlation between average house prices in Western Australia and average house prices in Australia as a whole has a correlation coefficient, r , equal to 0.83. The coefficient of determination, r^2 is then approximately 0.69. This means that approximately 69% of the variation in the average house price in Western Australia can be explained by the variation in the average house price in Australia as a whole.

Scattergraph of cooking times for joints of meat of various weights.



Using correlation coefficient, r , we might be tempted to interpret a linear association for which $r = 0.8$ as being twice as strong as an association for which $r = 0.4$. However, the r^2 values tell us that in the $r = 0.8$ association 64% of the variation in one variable can be explained by the association between the variables whilst in the $r = 0.4$ association just 16% is explained. Hence, in terms of the explained variation, the $r = 0.8$ association is four times the strength of the $r = 0.4$ association.

Example 2

Draw a scattergraph for the data shown below, determine r and r^2 , giving each value correct to two decimal places, and interpret the values obtained.

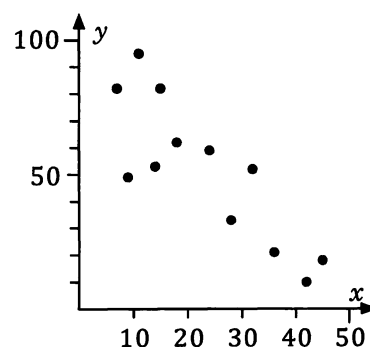
x	9	42	32	28	45	24	36	14	18	11	15	7
y	49	10	52	33	18	59	21	53	62	95	82	82

The scattergraph is shown on the right.

Using a calculator, $r = -0.84$ (correct to 2 dp),
and $r^2 = 0.71$ (correct to 2 dp).

The value of r informs us that x and y have a strong negative linear association.

The value of r^2 informs us that 71% of the variation in the variables is explained by the linear association between them.



Exercise 2C

1. For the bivariate data:

x	44	40	66	30	76	59	66	31	50	73	57	35
y	73	78	67	81	55	65	60	57	70	60	72	82

Determine the least squares regression line $y = ax + b$ and the correlation coefficient for the data with the outlier (a) included,
(b) excluded.

2. (a) Display the following bivariate data as a scatterplot with the variable p on the horizontal axis and the variable q on the vertical axis.

p	35	60	20	47	27	61	53	15	70	79	41	56	32	24	43	30	12	62	47	80
q	44	55	34	45	38	69	51	34	43	49	43	54	40	35	47	36	30	52	50	46

- (b) Crop the data so that you are considering the sixteen points that are more suited to linear regression and determine the least squares regression line equation $q = ap + b$ for this cropped data.
- (c) Use your regression line to predict a value for q for a p value of 38.
- (d) Why would it be unwise to use the regression line from part (b) to predict the q value for a p value of 75?

3. The tables shown below give the x -values and the linear regression residuals ($y - \hat{y}$) for six sets of bivariate data. ($\hat{y}$ being the y value predicted by the regression line.) Following the tables the scattergraphs for these same six sets of bivariate data are shown.

Match each table to its associated graph.

Table 1.

x	1	2	3	4	5	6	7	8	9	10
Residuals ($y - \hat{y}$)	6.0	2.1	0.2	-3.7	-3.6	-5.4	-3.3	-1.2	0.9	8.0

Table 2.

x	1	2	3	4	5	6	7	8	9	10
Residuals ($y - \hat{y}$)	0	0	0	0	0	0	0	0	0	0

Table 3.

x	1	2	3	4	5	6	7	8	9	10
Residuals ($y - \hat{y}$)	-3.4	1.3	2.1	-0.2	1.5	2.3	-1.0	-3.3	-0.5	1.2

Table 4.

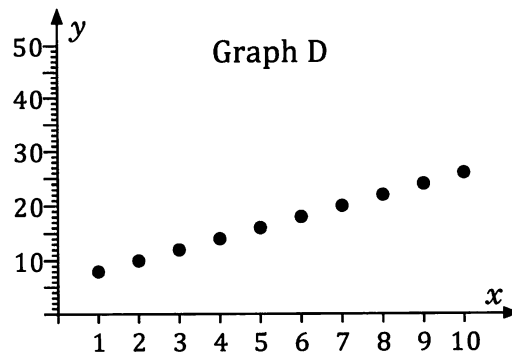
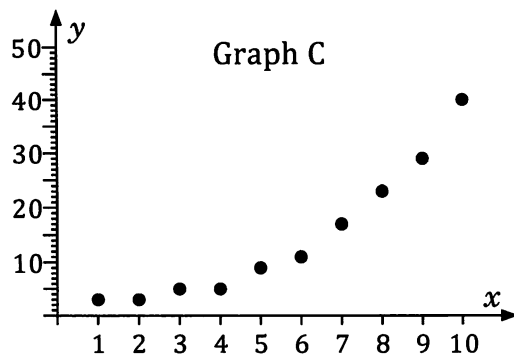
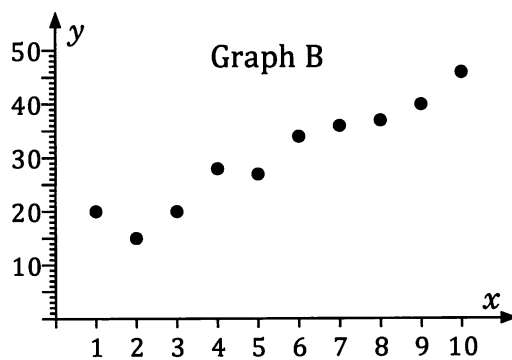
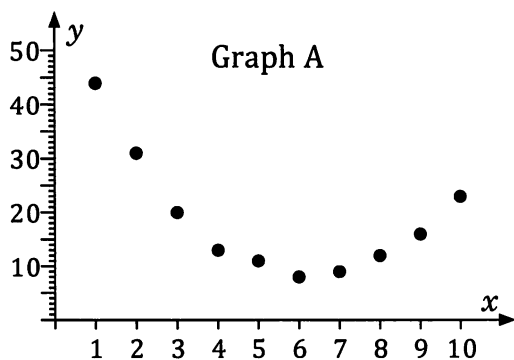
x	1	2	3	4	5	6	7	8	9	10
Residuals ($y - \hat{y}$)	15.8	4.9	-4.0	-8.9	-8.8	-9.6	-6.5	-1.4	4.7	13.8

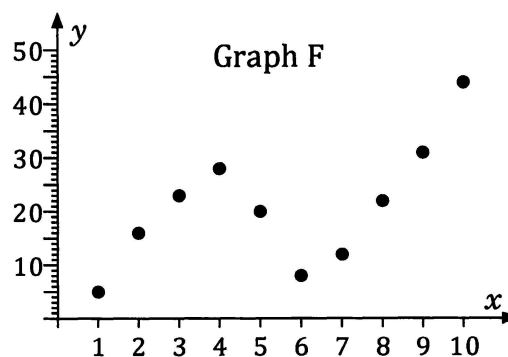
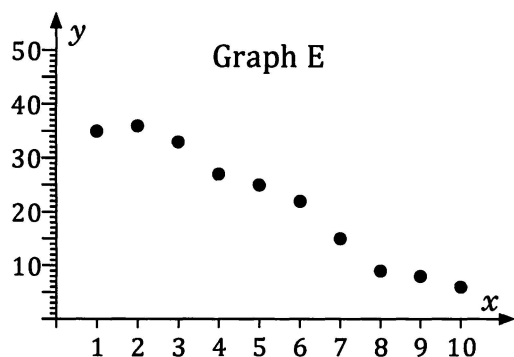
Table 5.

x	1	2	3	4	5	6	7	8	9	10
Residuals ($y - \hat{y}$)	-5.2	3.4	8.0	10.7	0.3	-14.1	-12.5	-4.8	1.8	12.4

Table 6.

x	1	2	3	4	5	6	7	8	9	10
Residuals ($y - \hat{y}$)	4.0	-4.2	-2.3	2.5	-1.7	2.1	0.9	-1.3	-1.4	1.4





4. An investigation into heights and weights of men of a particular age found that the correlation between height and weight had a correlation coefficient of 0.802. According to this result what percentage of the variation in the weight of men of this age can be explained by the variation in their heights?

5. A survey into the foot length of a number of primary school children and the reading age of these children showed a correlation between these two variables with a correlation coefficient, r , equal to 0.73.

According to this result what percentage of the variation in the reading age could be accounted for by the variation in foot length.

Does this mean that increasing the length of a person's foot causes their reading age to increase? Discuss.



6. Draw a scattergraph for the data shown below, determine r and r^2 , giving each value correct to two decimal places, and interpret the values obtained.

x	8	21	20	19	40	53	61	55	72	69
y	10	11	20	37	29	8	24	45	32	49

7. A study of the number of units of fertilizer initially added to the soil mix in seed trays in which seeds of a particular type are to be sown, and the average height of the seedlings in each tray six weeks after sowing, yielded the following results:

Units of fertiliser added initially.	0	1	2	3	4	5	6	7	8
Mean height of seedling after six weeks (in millimetres).	73	84	81	93	92	98	105	114	112

View the scattergraph of these results on a graphic calculator or computer and, if linear regression seems appropriate, determine what percentage of the variation in the height of the seedlings can be accounted for by the variation in the number of units of fertiliser added to the seed tray prior to planting.

8. Scientists monitor the concentration in the levels of a pollutant existing in a lake, following an industrial accident and the ongoing clean up.

The pollutant level, P parts per million, present t days after the monitoring commenced is as follows:

t	0	1	2	3	4	5	6	7	8	9	10
P	500	430	314	286	234	185	155	140	118	99	71

- (a) The least squares regression line for this data is $P \approx 431.45 - 40.25t$.

Use this rule to complete the following table of residuals, to the nearest unit and draw the residual plot.

(Residual = observed value - value predicted by regression line).

t days	0	1	2	3	4	5	6	7	8	9	10
Residual											

- (b) Explain why these residuals suggest that *linear* regression may not be the most appropriate.

9. Measurements of two variables, h and r , are recorded by scientists investigating possible links between these two variables.

The recorded measurements are shown in the following table.

h	1	2	3	4	5	6	7	8	9	10
r	15	85	195	345	540	770	1045	1360	1715	2110

- (a) The least squares regression line for this data is $r \approx 233h - 463$.

Use this rule to complete the following table of residuals, to the nearest unit.

(Residual = observed value - value predicted by regression line).

h	1	2	3	4	5	6	7	8	9	10
Residual										

- (b) Explain why these residuals suggest that *linear* regression may not be the most appropriate.

Extension: Quadratic regression.

- (c) Suppose instead that r and h are thought to be related by a rule that is of the form

$$r = ah^2 + bh + c.$$

This is now in the form of a *quadratic equation*, not a linear equation.

Using a calculator that can perform quadratic regression, determine values for a , b and c .

- (d) Use the quadratic rule from (c) to predict r when $h = 15$ commenting on the reliability of your prediction.

ANSCOMBE'S QUARTET*

For each of the four data sets shown below use the statistical facilities of your calculator to determine the following, correct to two decimal places.

- The mean of the x scores.
- The standard deviation of the x scores.
- The mean of the y scores.
- The standard deviation of the y scores.
- The coefficient of linear correlation, r_{xy} .
- The equation of the y on x linear regression line.

☞ Present your results neatly and comment on anything you notice.

☞ As you carry out the above also view the scattergraph for each data set and comment on the suitability of using linear regression techniques for each set of data.

Data Set	x	10.0	8.0	13.0	9.0	11.0	14.0	6.0	4.0	12.0	7.0	5.0
One	y	8.04	6.95	7.58	8.81	8.33	9.96	7.24	4.26	10.84	4.82	5.68

Data Set	x	10.0	8.0	13.0	9.0	11.0	14.0	6.0	4.0	12.0	7.0	5.0
Two	y	9.14	8.14	8.74	8.77	9.26	8.10	6.13	3.10	9.13	7.26	4.74

Data Set	x	10.0	8.0	13.0	9.0	11.0	14.0	6.0	4.0	12.0	7.0	5.0
Three	y	7.46	6.77	12.74	7.11	7.81	8.84	6.08	5.39	8.15	6.42	5.73

Data Set	x	8.0	8.0	8.0	8.0	8.0	8.0	8.0	19.0	8.0	8.0	8.0
Four	y	6.58	5.76	7.71	8.84	8.47	7.04	5.25	12.50	5.56	7.91	6.89

* F. J. Anscombe of Yale University presented these four data sets in an article published in *The American Statistician*, Volume 27, 1973. The sets are reproduced here with the permission of *The American Statistical Association*.

Data collection, display and analysis.

Collect data to investigate one of the questions listed below (or a similar sort of question of your own) and write a report of your investigations. Your report should include the following:

- ☞ The question you decided to investigate.
- ☞ What data you collected.
- ☞ How you collected the data.
- ☞ The data, tabulated and/or graphed as appropriate.
- ☞ Your conclusions and reasoning. (Including mention of any improvements that could be made to the investigation were it to be repeated.)

QUESTIONS

- Is there a relationship between the height of parents and the heights of their children?
- Is there a relationship between the age of the intended readership of a children's book and average sentence length?
- If a year 8 student is good at mathematics are they also likely to be good at English?
- Is there a relationship between height and hand span?
- Is the number of cold drinks sold in the school canteen during lunchtime related to the temperature of the day?

Miscellaneous Exercise Two.

This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

1. A study of a number of fires occurring in a metropolitan area found that there was a strong positive correlation between the number of firemen attending a fire and the amount of damage done by the fire. With this high positive correlation showing that

more firemen attending, more damage done

would we be correct to conclude that the firemen were causing the damage? Explain your answer.

2. Is there an association between whether or not someone is a regular cyclist and gender?

Let us suppose that to investigate the above question researchers defined what they meant by being a "regular cyclist" and then asked a number of people, all aged between 18 and 50, if they were regular cyclists. The results gave rise to the following table.

	Male	Female
Regular cyclist	52	19
Not a regular cyclist	132	102

Use these results to suggest an answer to the question posed at the beginning of this question and explain your reasoning.

3. In the course of an experiment scientists record values of two variable quantities, x and y , at a number of times. The paired values recorded were as follows:

x	51.2	52.2	53.1	50.6	52.3	51.6	50.9	52.8	51.7	50.2	50.1
y	31.3	26.5	26.2	36.7	30.7	31.5	34.2	25.1	31.3	36.3	37.8

Construct a scattergraph of these results and, purely by "eyeballing" your graph, estimate the value of the variable y

- (a) when x is 50.8,
 (b) when x is 53.2,
 and (c) when x is 58.

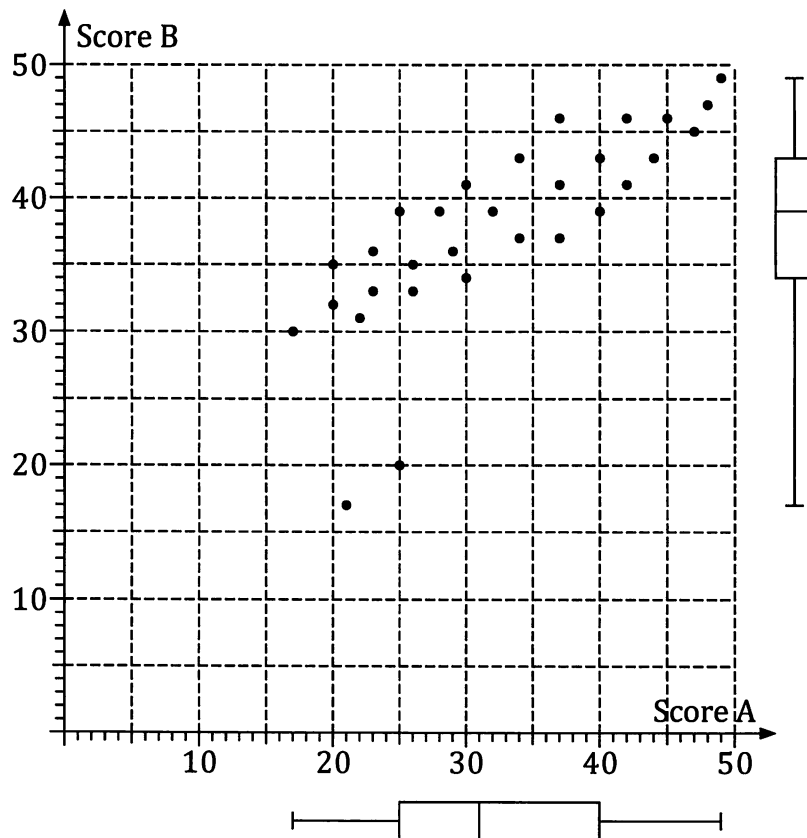
in each case commenting on the likely reliability of your prediction.

4. To pass a particular university unit students have to average at least 50% in three tests. If a student misses one of the tests, and has a valid reason for doing so, an estimated mark can be given. Sixteen students take the unit. Fifteen complete all three tests and student number 12 misses test C due to hospitalisation. The percentages achieved by the fifteen students sitting all three tests are given below:

Student N ^o .	1	2	3	4	5	6	7	8	9	10	11	13	14	15	16
Test A	30	86	36	66	40	44	76	10	33	74	23	53	77	16	55
Test B	60	91	41	57	45	60	70	12	22	80	40	36	71	9	86
Test C	40	76	50	88	70	62	70	40	60	79	26	75	94	53	63

- (a) By comparing correlation coefficients determine which of tests A and B is the better indicator of performance in test C for a linear model.
 (b) Use this "better indicator" to estimate a test C mark (to nearest 1%) for student 12 given that they scored 59% in test A and 48% in test B.
 (c) Which of the sixteen students do not pass the unit?

5. The scattergraph below is for the marks obtained by 30 students who sat two tests, A and B. The associated boxplots are also shown.



- (a) Compare the location and spread of the marks in the two tests.
- (b) The data is to be used to estimate a test B score for a 31st student who scored 27 in test A but missed test B. Determine an appropriate regression line, explaining your choice, state its equation, make the estimate and comment on the likely validity of your estimate.

In the first half of the 20th century the increasing death rate due to lung cancer in a number of countries was causing concern. Some people suggested that smoking was to blame. Some argued that whilst there might be an association between smoking and lung cancer the association was not necessarily causal.

More recently the debate has expanded to include consideration of "passive" smoking, the name given to the inhalation of "second-hand" smoke.

Use the internet to research this correlation / causation debate regarding smoking and lung cancer.